

NOTE TO FILE: NDE ON ROTOR AT DISASSEMBLY, TOGETHER WITH ANDRITZ ENGINEERING EVALUATION, DETERMINED THE NEED TO STRUCTURALLY MODIFY THE ROTOR (SEE BELOW).



Installation INSPECTION AND TEST PLAN

Document #: I-GMS - 6300 **Revision :** 03
Project: GMS Poles Unit 1-4 **Unit #:** 4

Codes: HC : hold point (customer)
 WC : witness point (customer)
 HA : hold point (Andritz Technical support)
 WA : witness point (Andritz Technical support)

ITP step #	Code	Description	Procedure #	Procedure step #	Comments	Form	Date and initials
Rotor inspection							
6300 - 1	WC	Rotor assembly inspection for fretting	P-6300-03	1.1		Q-0000-001	OK 2012.04.11
6300 - 2	WC	Rotor assembly general visual inspection	P-6300-03	1.2		Q-0000-001	OK 2012.04.11
Rotor inspection for possible cracks + ROTOR MODIFICATION (REWEELD RIBS + ADD GUSSETS)							
6300 - 3	WC	MT inspection on the following welds: - Rotor spider arms to the upper and lower disks - Other major weld susceptible to cracking	P-GMS-6300-99	2.3	With sub-contractor report	Q-0000-001	OK 2012.03.16
6300 - 3.1		Rib groove contour: - Visual inspection - MT inspections	P-GMS-6300-98	4.9		Q-0000-001 Q-0400-012	OK 2012.05.08
6300 - 3.2		Ribs backgouging - MT inspections	P-GMS-6300-98	5.13		Q-0400-012	OK 2012.05.21

Installation INSPECTION AND TEST PLAN

Document #: I-GMS - 6300 **Revision :** 03
Project: GMS Poles Unit 1-4 **Unit #:** 4

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ITP step #	Code	Description	Procedure #	Procedure step #	Comments	Form	Date and initials
6300 - 3.3		Rotor ribs welds: - Visual inspection - NDT inspections	P-GMS-6300-98	5.18		Q-0400-011 Q-0400-012	<i>OK</i> 2013-05-28
6300 - 3.4		Gusset plates welds: - Visual inspection - NDT inspections	P-GMS-6300-98	6.14		Q-0400-011 Q-0400-012	<i>OK</i> 2013-06-04
6300 - 4	WC	Rotor spider painting visual inspection	P-GMS-6300-99	2.4	New paint applied over inspected areas	Q-0600-010	<i>OK</i> 2013-06-20
6300 - 5	WC HA	Rotor rim offset inspection	P-GMS-6300-99	3.1 3.2		Q-6301-020C Q-0000-001	<i>OK</i> Sept 27, 12 <i>OK</i> 2013-07-16
6300 - 6	WC HA	Rotor rim skew inspection	P-GMS-6300-99	3.5 3.6		Q-GMS-6301-053 Q-0000-001	<i>OK</i> 2013-07-16

Installation INSPECTION AND TEST PLAN

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Unit #: 4

Project: GMS Poles Unit 1-4

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ITP step #	Code	Description	Procedure #	Procedure step #	Comments	Form	Date and initials
Rotor rim un-shrink and re-shrink for spider ledge inspection or rim measurements correction (if required)							
6300 - 7		Step cancelled					
6300 - 8		Step cancelled					
6300 - 9		Step cancelled					
6300 - 10	HC HA	- Rotor spider level - Rotor rim diameter, circularity, concentricity and verticality	P-GMS-6300-99	4.8	After rim shrink	Q-6301-020C	OK 2012-08-21
6300 - 11	WC	New poles stop blocks welding	P-GMS-6300-99	5.2	If necessary	Q-0400-011	OK 2012-08-21 Sept 27, 12

Installation INSPECTION AND TEST PLAN

Document #: I-GMS - 6300 **Rotor assembly** **Revision : 03**
Project: GMS Poles Unit 1-4 **Unit #: 4**

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ITP step #	Code	Description	Procedure #	Procedure step #	Comments	Form	Date and initials
Rotor cleaning							
6300 - 12	WC	Rotor cleaning visual inspection	P-6300-02A	1.4		Q-0000-001	2012-09-26 12-09-29
Rotor poles installation							
6300 - 13	HC	Megger test each pole at 1000Vdc	P-GMS-6301-04	1.3	Before installation	Q-0500-002	2012-08-23
6300 - 14	HC	Rotor Pole drop test	P-GMS-6325-03	1.4	Before installation	Q-6325-030	2012-09-27
6300 - 15	WC	Half length of each pole dovetail	P-GMS-6301-04	1.6		Q-GMS-6301-051	2012-08-22 12-09-29
6300 - 16	HC HA	Rotor/stator magnetic center level and elevation	P-GMS-6301-04	2.4		Q-GMS-6301-051	2012-08-22 12-09-29
6300 - 17	HC HA	Rotor diameter and poles circularity, concentricity and verticality	P-GMS-6301-04	2.6		Q-6301-041	2012-09-19 12-09-29

Installation **INSPECTION AND TEST PLAN**

Document #: I-GMS - 6300 **Revision :** 03
Project: GMS Poles Unit 1-4 **Unit #:** 4

Codes: HC : hold point (customer)
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ITP step #	Code	Description	Procedure #	Procedure step #	Comments	Form	Date and initials
6300 - 18	HC HA	- Megger test each pole individually at 5kVdc - Hipot test each pole at 5kVac or 7.5kVdc for 1 minute	P-GMS-6301-04	2.7	After installation	Q-0500-002	<i>[Signature]</i> 2013-09-04 <i>[Signature]</i> 12-09-14
6300 - 19	WC	Interpolar baffles torque	P-GMS-6301-04	2.8		Q-0000-003	<i>[Signature]</i> 2013-09-18 <i>[Signature]</i> 12-09-14
6300 - 20	WC	Rotor connection radial gaps	P-GMS-6301-04	2.9		Q-6323-010	<i>[Signature]</i> 2013-09-18 <i>[Signature]</i> 12-09-14
6300 - 21	WC	Pole-to-pole connection torque	P-GMS-6301-04	2.10		Q-0000-003	<i>[Signature]</i> 2013-09-18 <i>[Signature]</i> 12-09-14
6300 - 22	WC	Rotor leads torque	P-GMS-6301-04	2.11		Q-0000-003	<i>[Signature]</i> 2013-09-18 <i>[Signature]</i> 12-09-14
6300 - 23	HC HA	Visual inspection of the rotor prior to testing, Megger test at 5kVdc and hipot test the on the entire rotor at 5kVac or 7.5kVdc for 1 minute	P-GMS-6301-04	2.14	Poles connected	Q-0500-001	<i>[Signature]</i> 2013-09-25



Installation INSPECTION AND TEST PLAN

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ITP step #	Code	Description	Procedure #	Procedure step #	Comments	Form	Date and initials
6300 - 24	HC	Rotor pole drop test	P-GMS-6325-03	1.4	Poles connected	Q-6325-030	<i>[Signature]</i> 2012-09-24
6300 - 25	HC	Rotor resistance test	P-GMS-6301-04	2.16	TO BE COMPLETED DURING COMMISSIONING <i>[Signature]</i> 2012-09-24	Q-1510-031	<i>[Signature]</i> 2012-09-24
6300 - 26	HC	Final inspection				Q-0000-002	<i>[Signature]</i> 2012-09-24



Installation INSPECTION AND TEST PLAN

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ITP step #	Code	Description	Procedure #	Procedure step #	Comments	Form	Date and initials
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Revision summary

Revision 03

Prepared by: François Sergerie
Approved by: François Sergerie

Date: 16-Apr-2012

- Step 6300-03.1 modified;
- Step 6300-13 modified;

Revision 02

Prepared by: François Sergerie
Approved by: François Boulet

Date: 5-Mar-2012

- Step 6300-03 modified;
- Step 6300-03.1 added;
- Step 6300-03.2 added;
- Step 6300-03.3 added;
- Step 6300-03.4 added;
- Step 6300-04 modified;
- Step 6300-07 cancelled;
- Step 6300-08 cancelled;
- Step 6300-09 cancelled;
- Step 6300-10 modified;
- Step 6300-11 modified;



Installation INSPECTION AND TEST PLAN

Document #: I-GMS - 6300 **Revision :** 03
Project: GMS Poles Unit 1-4 **Unit #:** 4

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ITP step #	Code	Description	Procedure #	Procedure step #	Comments	Form	Date and initials
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Revision 01

Prepared by: François Sergerie
Approved by: François Boulet

Date: 1-Feb-2012

- Step 6300-05 modified.

Revision 00

Prepared by: François Sergerie
Approved by: François Boulet

Date: 8-Nov-2011



Installation and Commissioning
Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 6300-1

Revision # 00

Project: GMS

Unit : G4

Page 1 of 1

Equipment : Rotor Assembly Inspection for fretting

At the time of mobilisation, the rotor was on the pedestal on the service bay.
The rotor was therefore cleaned up prior to inspection. No evidence of fretting
could be observed inside the rotor. However there were some evidence of fretting during
the inspection outage of 2011 (as documented in ITP 6000-6)

NCR #:

Measurement Instruments	T°	Measured by:	dd-mmm-yyyy
Visual	Before:		
	After:	ANDRITZ Rep.: Jean-Luc Cote	11-Apr-2012
		Cust. Repres.	



Installation and Commissioning
Quality Control Record

General Inspection Report

Q-0000-001

Revision # 00

Page 1 of 1

ITP step #: 6300-2

Unit : 4

Project: GMS

Equipment : Rotor Assembly General Visual Inspection

The rotor was visually inspected. The only difference between the inspection outage was that we had access to the poles. The existing poles did not appear in good condition. The bottom face of the coils were covered in black dust (likely carbon powder from the brakes). An additional remark was that the rotor rim driven key welds were cracked (expected considering the rim becoming loose at rather low speed). Another observation was that the rotor rim cross braces (located close to the rotor rim) appeared to have fresh paint touch-ups judging by the color.

NCR #: _____

Measurement Instruments	T°	Measured by: _____	dd-mmm-yyyy
Visual	Before: _____	ANDRITZ Rep.: Jean-Luc Cote	11-Apr-2012
	After: _____	Cust. Repres. _____	



Installation and Commissioning
Quality Control Record

General Inspection Report

Q-0000-001

ITP step #: 6300-3 & 3.1

Revision # 00

Project: GMS

Unit : 4

Page 1 of 1

Equipment : As-found rotor spider MT inspection

The rotor spider arms to the upper and lower disks, the rib contour, and the cross braces
were inspected by CWB certified third party level 2 MT inspectors (Radius NDE Inspection)

Please see attached report for indications observed

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Dean Beamish</u>	<u>dd-mmm-yyyy</u>
	Before: _____	<u>Wayne Fuhr</u>	
	After: _____	ANDRITZ Rep.: <u>Jean-Luc Cote</u>	<u>May 8th 2012</u>
		Cust. Repres. _____	

9004 - 107TH STREET
FORT ST. JOHN BC
V1J 5R5

RADIUS

NDE SERVICES INC.

PHONE (250) 787- 0336
FAX (250) 787- 0733

May 09, 2012

Andritz Hydro Ltd
6100 aut Transcanadienne
Pointe-Claire Quebec H9R 1B9

Reference Daily Reports
14335, 14336, 14337,
14338 and 14339

Attention : Jean-Luc Cote

RE: Inspection of G-4 Cross Braces and Spider Assembly Welds - WAC Bennett Dam

Dear Mr. Cote:

On May 1, 2, 4, 7 and 8th two Radius NDE Services Inc. Inspectors attended the WAC Bennett / GM Shrum Generating Station at your request. The purpose of the site visit was to conduct the magnetic particle inspection of all upper and lower welded joints on 20 cross braces, 76 plus inches of all upper and lower longitudinal welded joints in 20 rotor pockets, 20 plug welds and the bottom disk to the coupling ring weld of the unit G-4 Rotor. The primary basis of the inspection was to detect and identify crack indications, as a result, all pre-existing welding defects such as porosity and undercutting were not considered.

The inspection consisted of wet, colour contrast, multi-directional A/C magnetic particle scans of the welded joints and approximately one inch of parent material on either side of the weld in accordance with standard industry practice.

Several crack like indications were identified during the course of the inspection. These areas were demonstrated to onsite Hydro staff for rework and consisted of four indications on longitudinal Pocket joints, one area of light indications in Pocket #1, one indication on a lower Cross Brace and three indications on the inner Spider Arms. No indications were noted on any of the 20 Plug welds and the Bottom Disk to Coupling Ring weld.

Please see the following graph and photo pages.

Sincerely;

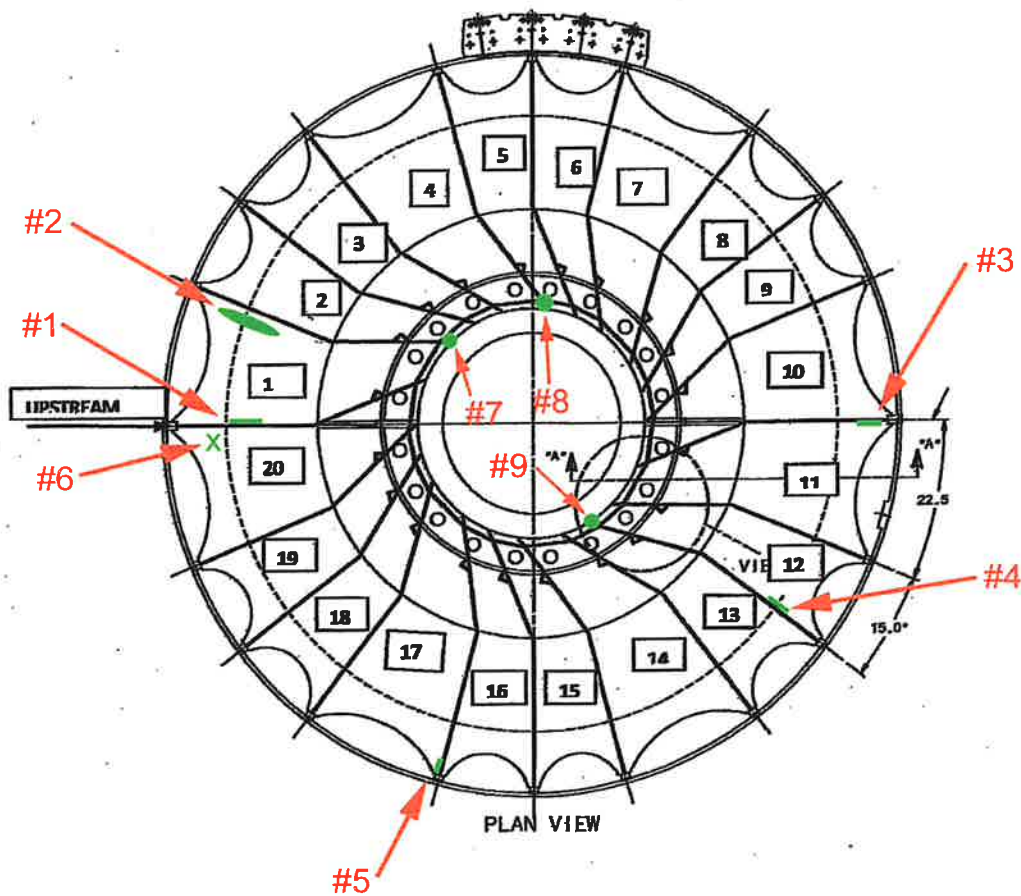


Wayne Fuhr - CGSB Level II MT, PT
Registration # 12707

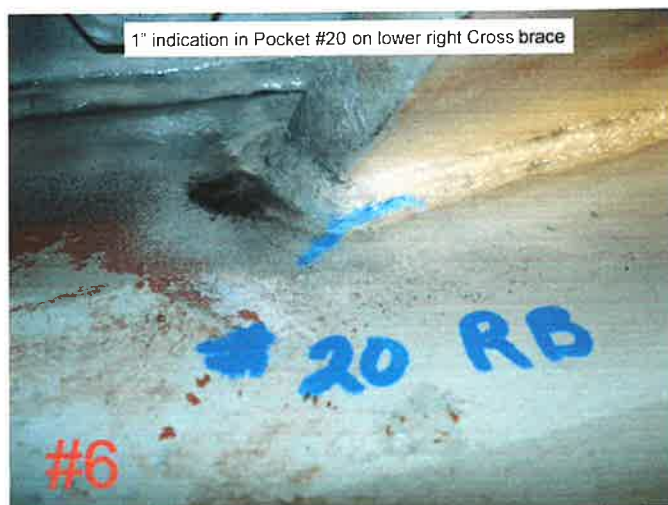
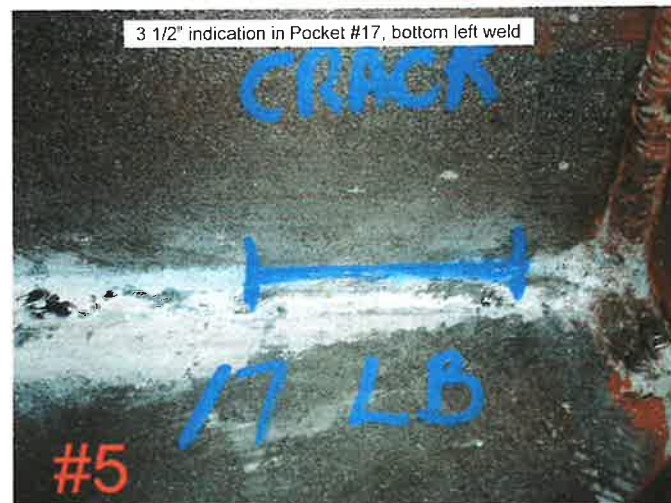
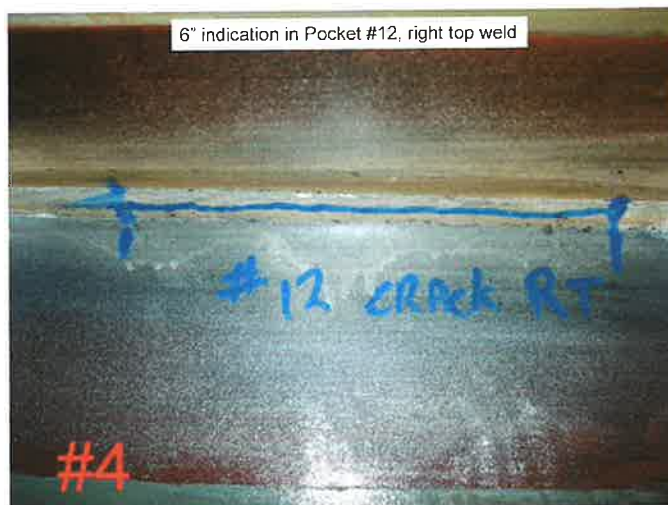
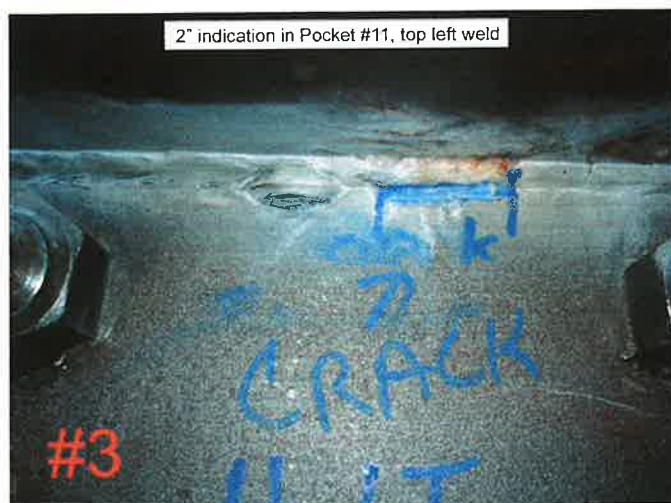
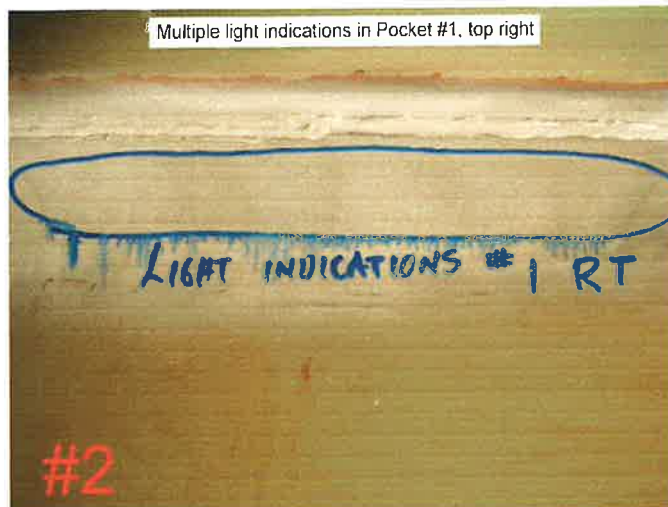
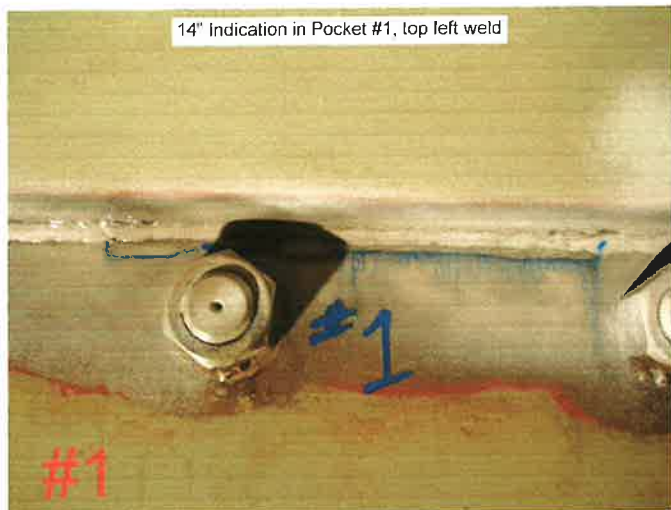


Dean Beamish - CGSB Level II RT, MT, PT
Registration # 5380

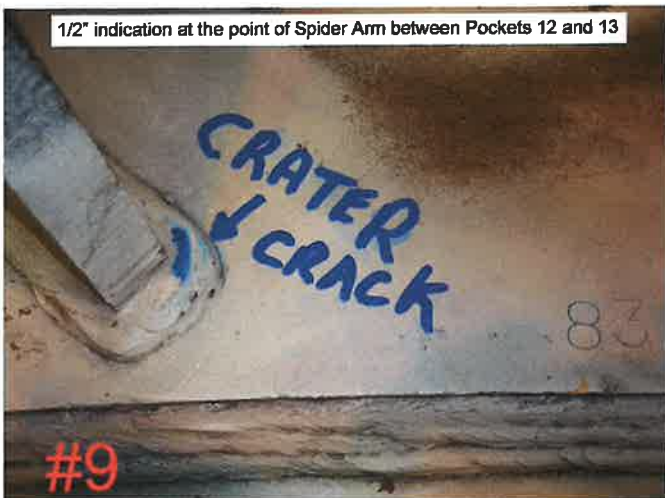
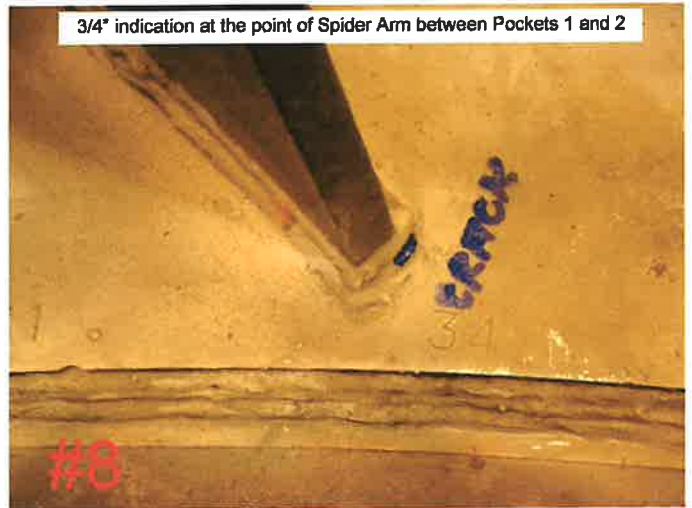
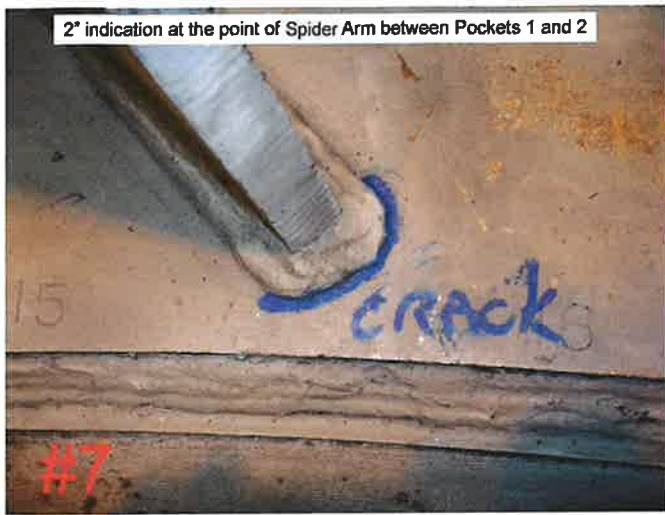
GMS ROTOR POCKETS UNIT #4



Photographs



Photographs



Installation and Commissioning Quality Control Record

Non-Destructive Test

Q-0400-012

 ITP step #: 6300-3.1

Revision # 00

 Project: GMS

 Unit : G4

Page 1 of 3

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
#1 Pocket Left Side	6300-3.1	X				X	25-05-12	LM
#1 Pocket Right Side	6300-3.1	X					25-05-12	LM
#2 Pocket Left Side	6300-3.1	X					25-05-12	LM
#2 Pocket Right Side	6300-3.1	X					25-05-12	LM
#3 Pocket Left Side	6300-3.1	X					25-05-12	LM
#3 Pocket Right Side	6300-3.1	X					25-05-12	LM
#4 Pocket Left Side	6300-3.1	X					25-05-12	LM
#4 Pocket Right Side	6300-3.1	X					25-05-12	LM
#5 Pocket Left Side	6300-3.1	X					25-05-12	LM
#5 Pocket Right Side	6300-3.1	X					25-05-12	LM
#6 Pocket Left Side	6300-3.1	X				X	25-05-12	LM
#6 Pocket Right Side	6300-3.1	X					25-05-12	LM
#7 Pocket Left Side	6300-3.1	X					25-05-12	LM
#7 Pocket Right Side	6300-3.1	X					25-05-12	LM
CONTINUED NEXT PAGE								

 Comments : #1 Pocket left side, and pocket #6 Left side revealed indications.

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>May 25th 2012</u>
		Cust. Repres. _____	

Installation and Commissioning Quality Control Record

Non-Destructive Test

Q-0400-012

ITP step #: 6300-3.1

Revision # 00

Project: GMS

Unit : G4

Page 2 of 3

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
#8 Pocket Left Side	6300-3.1	X					25-05-12	LM
#8 Pocket Right Side	6300-3.1	X				X	25-05-12	LM
#9 Pocket Left Side	6300-3.1	X					25-05-12	LM
#9 Pocket Right Side	6300-3.1	X					25-05-12	LM
#10 Pocket Left Side	6300-3.1	X				X	25-05-12	LM
#10 Pocket Right Side	6300-3.1	X					25-05-12	LM
#11 Pocket Left Side	6300-3.1	X					25-05-12	LM
#11 Pocket Right Side	6300-3.1	X					25-05-12	LM
#12 Pocket Left Side	6300-3.1	X					25-05-12	LM
#12 Pocket Right Side	6300-3.1	X					25-05-12	LM
#13 Pocket Left Side	6300-3.1	X					25-05-12	LM
#13 Pocket Right Side	6300-3.1	X				X	25-05-12	LM
#14 Pocket Left Side	6300-3.1	X					25-05-12	LM
#14 Pocket Right Side	6300-3.1	X					25-05-12	LM
CONTINUED NEXT PAGE								

Comments : #8 Pocket right side, #10 pocket left side, #13 pocket right side revealed indications

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>May 25th 2012</u>
		Cust. Repres. _____	_____

Installation and Commissioning Quality Control Record

Non-Destructive Test

Q-0400-012

ITP step #: 6300-3.1

Revision # 00

Project: GMS

Unit : G4

Page 3 of 3

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
#15 Pocket Left Side	6300-3.1	X				X	25-05-12	LM
#15 Pocket Right Side	6300-3.1	X					25-05-12	LM
#16 Pocket Left Side	6300-3.1	X					25-05-12	LM
#16 Pocket Right Side	6300-3.1	X				X	25-05-12	LM
#17 Pocket Left Side	6300-3.1	X					25-05-12	LM
#17 Pocket Right Side	6300-3.1	X					25-05-12	LM
#18 Pocket Left Side	6300-3.1	X					25-05-12	LM
#18 Pocket Right Side	6300-3.1	X					25-05-12	LM
#19 Pocket Left Side	6300-3.1	X					25-05-12	LM
#19 Pocket Right Side	6300-3.1	X				X	25-05-12	LM
#20 Pocket Left Side	6300-3.1	X					25-05-12	LM
#20 Pocket Right Side	6300-3.1	X					25-05-12	LM

Comments : #15 pocket left side, #16 pocket right side, #19 pocket right side
revealed indications.

All indications were repaired and re-tested to ensure complete removal

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>May 25th 2012</u>
		Cust. Repres. _____	

RADIUS

NDE SERVICES INC.

9004 – 107th Street
Fort St. John, BC V1J 5R5
(250) 787-0336 phone
(250) 787-0733 fax

Client:	Andritz Hydro Ltd	Date	12.05.21
Attention:	Jean-Luc Cote	Page	1 of 2
Charge Code:	Unit #4 Refurbishment	Ticket #:	14340
Location:	GM Shrum Generating Station		

MAGNETIC PARTICLE INSPECTION REPORT

TEST METHOD:	Colour Contrast / A/C Yoke
ACCEPTANCE STANDARD:	Client Spec / Crack Detection
OBJECT:	Inner 10" of Spider Arms, post gouging operations

TEST CONDITIONS

EQUIPMENT:	MANUFACTURER: Electro Spect				TYPE: ES-X	
MAGNETIC PARTICLES:	MANUFACTURER: Magnaflux				TYPE: 7HF	
	☐ DRY	☒ WET	☐ FLUORESCENT	☒ NON-FLUORESCENT COLOUR:	Black	
MAGNETIC METHOD:	☐ PROD	☒ YOKE	☐ COIL	☒ CONTINUOUS	☐ RESIDUAL	☒ AC ☐ DC
SURFACE CONDITION:	☒ AS GROUND	☐ MACHINED	☐ SHOT BLASTED	☐ BASE METAL	☐ PAINTED SURFACE	☐ Buffed

EXAMINATION RESULTS:

Inspected both sides of the inner 10" on the Spider Arms post welding on one side and gouging operations on the other side. The inspection consisted of wet, colour contrast, multi-directional A/C magnetic particle scans of the welded joints and approximately one inch of parent material on either side of the weld area in accordance with standard industry practice.

Eight crack like indications were identified during the course of the inspection. These areas were demonstrated to onsite Andritz staff for rework and consisted of indications in four areas on various Spider Arms in the gouged side and four other areas on the welded side of four other Spider Arms. All indications were removed by grinding and retested to ensure complete removal. No other indications were noted at this time.

Please see the following photograph page.

Three indications on the Spider Arms identified in photographs #7, 8 and 9 of the report dated May 09, 2012 were removed by gouging and grinding operations.

W. Fuhr



II

12707

TECHNICIAN

SIGNATURE

LEVEL

REGISTRATION #

The above examination results are the professional opinion of the Technician, but in no way guarantee the integrity of the product.

D. Beamish



II

5380

TECHNICIAN

SIGNATURE

LEVEL

REGISTRATION #

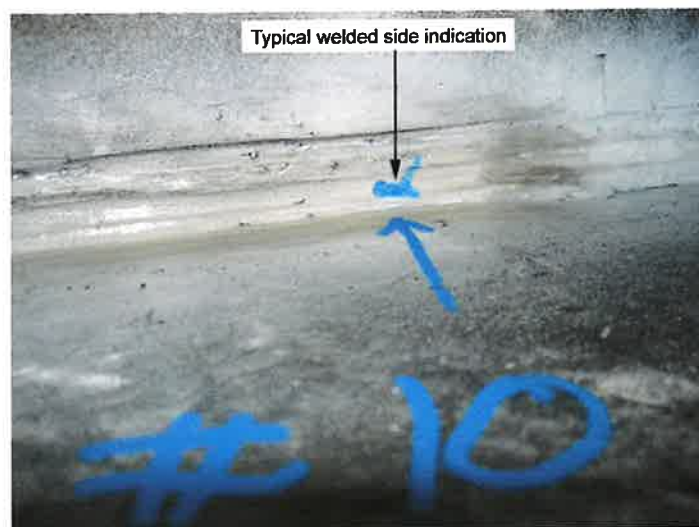
RADIUS

NDE SERVICES INC.

9004 – 107th Street
Fort St. John, BC V1J 5R5
(250) 787-0336 phone
(250) 787-0733 fax

Client:	Andritz Hydro Ltd	Date:	12.05.21
Attention:	Jean-Luc Cote	Page	2 of 2
Charge Code:	Unit #4 Refurbishment	Ticket #:	14340
Location:	GM Shrum Generating Station		

PHOTOGRAPHS



Installation and Commissioning **Quality Control Record**

Non-Destructive Test

Q-0400-012

ITP step #: 6300-3.2

Revision # 00

Project: GMS

Unit : G4

Page 1 of 3

Backgouging inspection
of rotor ribs

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Rotor Ribs Backgouging	6300-3.2	X				X	21-05-12	LM

Comments : 4 areas on the backgouge side and 4 areas on the welded sides
revealed indications. All indications were removed by grinding and
retested to ensure complete removal. No other indications were noted

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	<u>dd-mmm-yyyy</u>
<u>Magnetic Particles</u>	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>May 21st 2012</u>
		Cust. Repres. _____	

RADIUS

NDE SERVICES INC.

9004 – 107th Street
Fort St. John, BC V1J 5R5
(250) 787-0336 phone
(250) 787-0733 fax

Client:	Andritz Hydro Ltd	Date	12.05.21
Attention:	Jean-Luc Cote	Page	1 of 2
Charge Code:	Unit #4 Refurbishment	Ticket #:	14340
Location:	GM Shrum Generating Station		

MAGNETIC PARTICLE INSPECTION REPORT

TEST METHOD:	Colour Contrast / A/C Yoke
ACCEPTANCE STANDARD:	Client Spec / Crack Detection
OBJECT:	Inner 10" of Spider Arms, post gouging operations

TEST CONDITIONS

EQUIPMENT:	MANUFACTURER: Electro Spect				TYPE: ES-X	
MAGNETIC PARTICLES:	MANUFACTURER: Magnaflux				TYPE: 7HF	
	☐ DRY	☒ WET	☐ FLUORESCENT	☒ NON-FLUORESCENT COLOUR:		Black
MAGNETIC METHOD:	☐ PROD	☒ YOKE	☐ COIL	☒ CONTINUOUS	☐ RESIDUAL	☒ AC ☐ DC
SURFACE CONDITION:	☒ AS GROUND	☐ MACHINED	☐ SHOT BLASTED	☐ BASE METAL	☐ PAINTED SURFACE	☐ Buffed

EXAMINATION RESULTS:

Inspected both sides of the inner 10" on the Spider Arms post welding on one side and gouging operations on the other side. The inspection consisted of wet, colour contrast, multi-directional A/C magnetic particle scans of the welded joints and approximately one inch of parent material on either side of the weld area in accordance with standard industry practice.

Eight crack like indications were identified during the course of the inspection. These areas were demonstrated to onsite Andritz staff for rework and consisted of indications in four areas on various Spider Arms in the gouged side and four other areas on the welded side of four other Spider Arms. All indications were removed by grinding and retested to ensure complete removal. No other indications were noted at this time.

Please see the following photograph page.

Three indications on the Spider Arms identified in photographs #7, 8 and 9 of the report dated May 09, 2012 were removed by gouging and grinding operations.

W. Fuhr



II

12707

TECHNICIAN

SIGNATURE

LEVEL

REGISTRATION #

The above examination results are the professional opinion of the Technician, but in no way guarantee the integrity of the product.

D. Beamish



II

5380

TECHNICIAN

SIGNATURE

LEVEL

REGISTRATION #

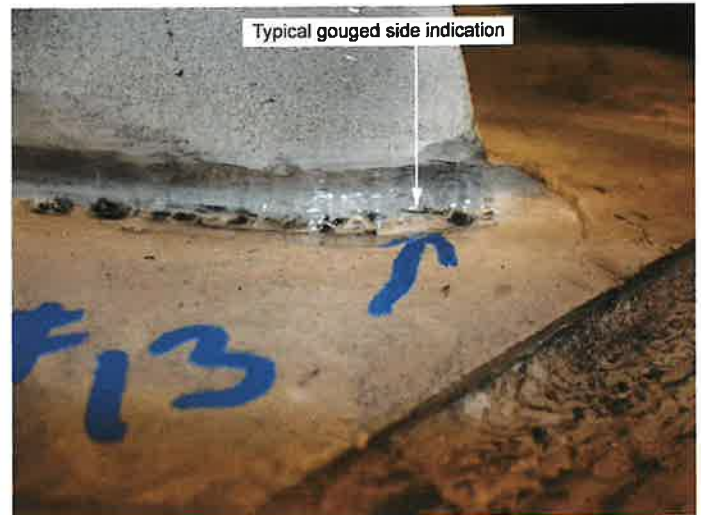
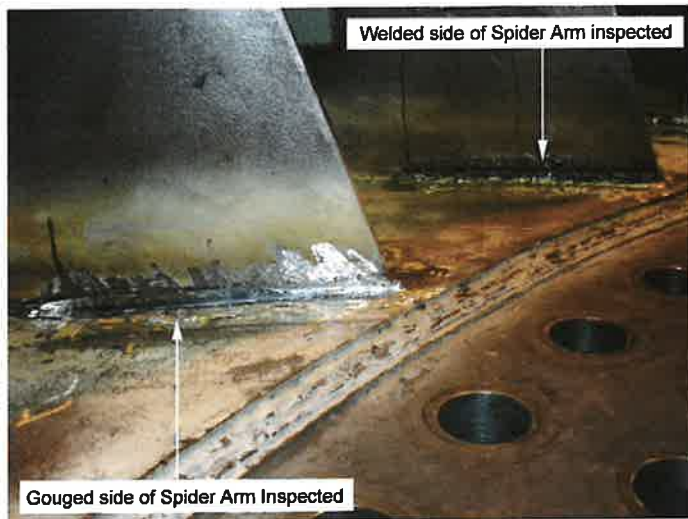
RADIUS

NDE SERVICES INC.

9004 – 107th Street
Fort St. John, BC V1J 5R5
(250) 787-0336 phone
(250) 787-0733 fax

Client:	Andritz Hydro Ltd	Date:	12.05.21
Attention:	Jean-Luc Cote	Page	2 of 2
Charge Code:	Unit #4 Refurbishment	Ticket #:	14340
Location:	GM Shrum Generating Station		

PHOTOGRAPHS



Installation and Commissioning Quality Control Record

Non-Destructive Test

Q-0400-012

ITP step #: 6300-3.3

Revision # 00

Project: GMS

Unit : G4

Page 1 of 3

Rotor Rib inspection
after welding

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Pocket #1 Left	6300-3.3	X					28-05-12	LM
Pocket #1 Right	6300-3.3	x					28-05-12	LM
Pocket #2 Left	6300-3.3	x					28-05-12	LM
Pocket #2 Right	6300-3.3	x					28-05-12	LM
Pocket #3 Left	6300-3.3	x					28-05-12	LM
Pocket #3 Right	6300-3.3	x					28-05-12	LM
Pocket #4 Left	6300-3.3	x					28-05-12	LM
Pocket #4 Right	6300-3.3	x					28-05-12	LM
Pocket #5 Left	6300-3.3	x					28-05-12	LM
Pocket #5 Right	6300-3.3	x					28-05-12	LM
Pocket #6 Left	6300-3.3	x					28-05-12	LM
Pocket #6 Right	6300-3.3	x					28-05-12	LM
Pocket #7 Left	6300-3.3	x				x	28-05-12	LM
Pocket #7 Right	6300-3.3	x				x	28-05-12	LM
CONTINUED NEXT PAGE								

Comments : Pocket #7 left and right, pocket #14 right and pocket #18 left
revealed indications. All indications were removed by grinding and
retested to ensure complete removal. No other indications were noted

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>May 28th 2012</u>
		Cust. Repres. _____	

Installation and Commissioning Quality Control Record

Non-Destructive Test

Q-0400-012

ITP step #: 6300-3.3

Revision # 00

Project: GMS

Unit : G4

Page 2 of 3

Rotor Rib inspection
after welding

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Pocket #8 Left	6300-3.3	X					28-05-12	LM
Pocket #8 Right	6300-3.3	x					28-05-12	LM
Pocket #9 Left	6300-3.3	x					28-05-12	LM
Pocket #9 Right	6300-3.3	x					28-05-12	LM
Pocket #10 Left	6300-3.3	x					28-05-12	LM
Pocket #10 Right	6300-3.3	x					28-05-12	LM
Pocket #11 Left	6300-3.3	x					28-05-12	LM
Pocket #11 Right	6300-3.3	x					28-05-12	LM
Pocket #12 Left	6300-3.3	x					28-05-12	LM
Pocket #12 Right	6300-3.3	x					28-05-12	LM
Pocket #13 Left	6300-3.3	x					28-05-12	LM
Pocket #13 Right	6300-3.3	x					28-05-12	LM
Pocket #14 Left	6300-3.3	x					28-05-12	LM
Pocket #14 Right	6300-3.3	x				x	28-05-12	LM
CONTINUED NEXT PAGE								

Comments : Pocket #7 left and right, pocket #14 right and pocket #18 left
revealed indications. All indications were removed by grinding and
retested to ensure complete removal. No other indications were noted

NCR #: _____

Measurement Instruments	T°	Measured by: Wayne Fuhr	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: Labib Mansour	May 28th 2012
		Cust. Repres. _____	

Installation and Commissioning Quality Control Record

Non-Destructive Test

Q-0400-012

 ITP step #: 6300-3.3

Revision # 00

 Project: GMS

 Unit : G4

Page 3 of 3

 Rotor Rib inspection
 after welding

Welding joint or part inspected	ITP step #	Test type				Default detected	Date (dd-mmm-yyyy)	Initials
		Magnetic particles	Liquid penetrant	Ultrasonic	Radiographic			
Pocket #15 Left	6300-3.3	X					28-05-12	LM
Pocket #15 Right	6300-3.3	x					28-05-12	LM
Pocket #16 Left	6300-3.3	x					28-05-12	LM
Pocket #16 Right	6300-3.3	x					28-05-12	LM
Pocket #17 Left	6300-3.3	x					28-05-12	LM
Pocket #17 Right	6300-3.3	x					28-05-12	LM
Pocket #18 Left	6300-3.3	x				x	28-05-12	LM
Pocket #18 Right	6300-3.3	x					28-05-12	LM
Pocket #19 Left	6300-3.3	x					28-05-12	LM
Pocket #19 Right	6300-3.3	x					28-05-12	LM
Pocket #20 Left	6300-3.3	x					28-05-12	LM
Pocket #20 Right	6300-3.3	x					28-05-12	LM

Comments : Pocket #7 left and right, pocket #14 right and pocket #18 left
revealed indications. All indications were removed by grinding and
retested to ensure complete removal. No other indications were noted

NCR #: _____

Measurement Instruments	T°	Measured by: <u>Wayne Fuhr</u>	dd-mmm-yyyy
Magnetic Particles	Before: _____		
	After: _____	ANDRITZ Rep.: <u>Labib Mansour</u>	<u>May 28th 2012</u>
		Cust. Repres. _____	

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NDE SERVICES INC.

9004 – 107th Street
Fort St. John, BC V1J 5R5
(250) 787-0336 phone
(250) 787-0733 fax

Client:	Andritz Hydro Ltd	Date	12.05.28
Attention:	Jean-Luc Cote	Page	1 of 3
Charge Code:	Unit #4 Refurbishment	Ticket #:	14341
Location:	GM Shrum Generating Station		

MAGNETIC PARTICLE INSPECTION REPORT

TEST METHOD:	Colour Contrast / A/C Yoke
ACCEPTANCE STANDARD:	Client Spec / Crack Detection
OBJECT:	Inner 10" of Spider Arms, post gouging operations

TEST CONDITIONS

EQUIPMENT:	MANUFACTURER: Electro Spect				TYPE: ES-X	
MAGNETIC PARTICLES:	MANUFACTURER: Magnaflux				TYPE: 7HF	
	☐ DRY	☒ WET	☐ FLUORESCENT	☒ NON-FLUORESCENT COLOUR: Black		
MAGNETIC METHOD:	☐ PROD	☒ YOKE	☐ COIL	☒ CONTINUOUS	☐ RESIDUAL	☒ AC ☐ DC
SURFACE CONDITION:	☒ AS GROUND	☐ MACHINED	☐ SHOT BLASTED	☐ BASE METAL	☐ PAINTED SURFACE	☐ Buffed

EXAMINATION RESULTS:

Inspected both sides of the inner 10" on the Spider Arms post welding operations on both sides. The inspection consisted of wet, colour contrast, multi-directional A/C magnetic particle scans of the welded joints and approximately one inch of parent material on either side of the weld area in accordance with standard industry practice.

Four crack like indications were identified during the course of the inspection. These areas were demonstrated to onsite Andritz staff for rework and consisted of two indications which were removed by grinding only and the other two indications required grinding and re-welding. All indications were retested to ensure complete removal. No other indications were noted at this time.

Please see the following chart and photograph pages.

W. Fuhr



II

12707

TECHNICIAN

SIGNATURE

LEVEL

REGISTRATION #

D. Beamish



II

5380

TECHNICIAN

SIGNATURE

LEVEL

REGISTRATION #

The above examination results are the professional opinion of the Technician, but in no way guarantee the integrity of the product.

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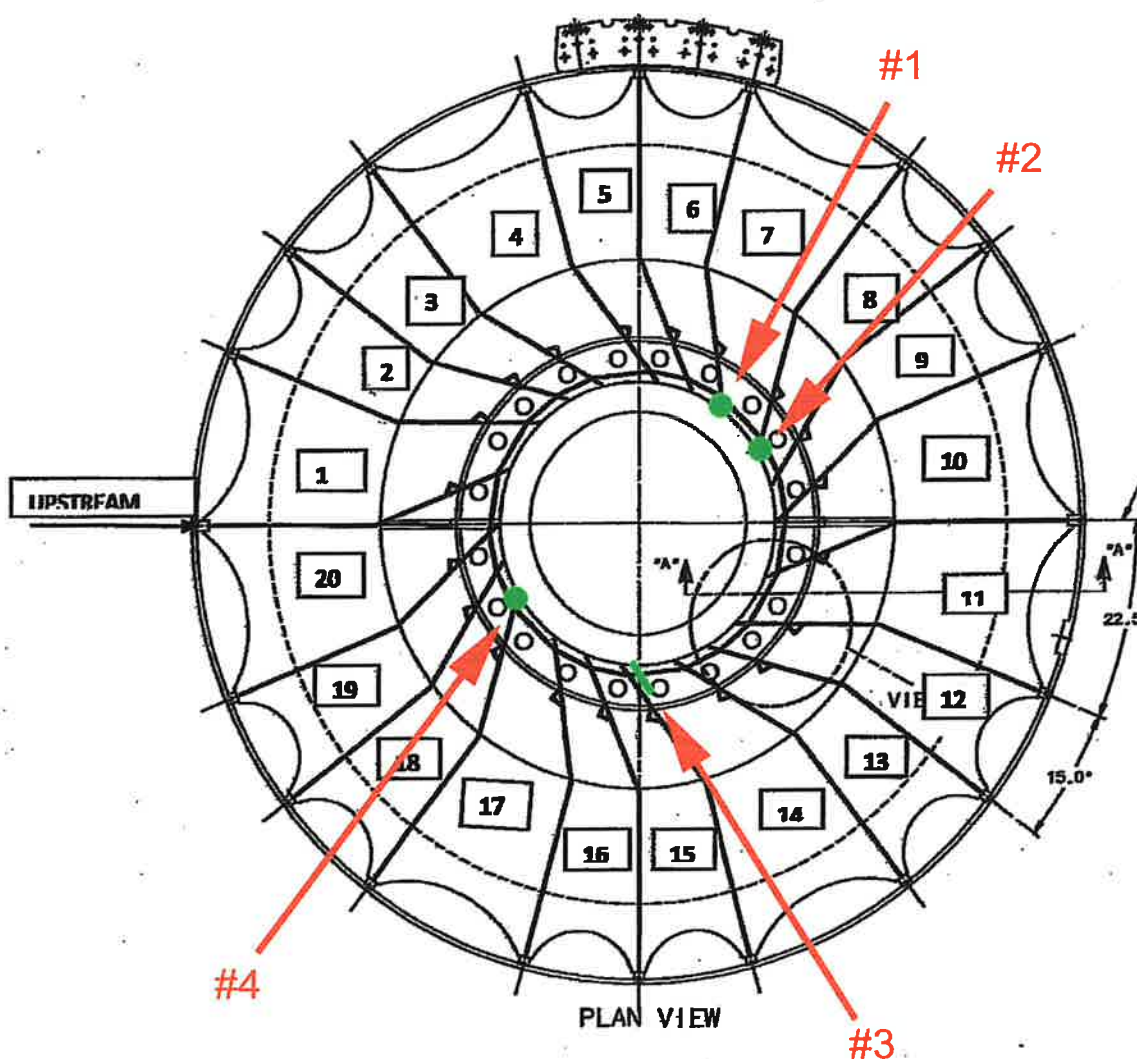
NDE SERVICES INC.

9004 - 107th Street
Fort St. John, BC V1J 5R5
(250) 787-0336 phone
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Client:	Andritz Hydro Ltd	Date:	12.05.28
Attention:	Jean-Luc Cote	Page 2 of	3
Charge Code:	Unit #4 Refurbishment	Ticket #:	14341
Location:	GM Shrum Generating Station		

CHART

GMS ROTOR POCKETS UNIT #4



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Fort St. John, BC V1J 5R5
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Client:	Andritz Hydro Ltd	Date:	12.05.28
Attention:	Jean-Luc Cote	Page	3 of 3
Charge Code:	Unit #4 Refurbishment	Ticket #:	14341
Location:	GM Shrum Generating Station		

PHOTOGRAPHS

